batch normalization

1. $Y=f(X)$, $X\in R^{d\times m}$, m examples with d features

$\mu=\frac{1}{m}\sum X_i$, $X=(X_1,X_2,X_i,X_m)$, i indexes examples.

$\sigma^2=\frac{1}{m}\sum(X_i-\mu)^2$, $\mu\in R^d$, $\sigma^2\in R^d$

$\hat{X}_i=\frac{X_i-\mu}{\sqrt{\sigma^2+\varepsilon}}$, $\varepsilon=10^{-6}$

$Y_i=k\hat{X}_i+b$, $k,b\in R^d$, $Y=(Y_1,Y_2,Y_i,Y_m)$,

2. $\frac{dL}{dY}$ is known, $L=L(Y)\in R\Rightarrow \frac{dL}{dX},\frac{dL}{dk},\frac{dL}{db}$

$\left(\frac{dL}{dX}:=\left(\frac{\partial L}{\partial X}\right).\text{reshape}(X.\text{shape}),\ \forall \text{ tensor } X\right)$

solve:

① $\frac{dL}{d\hat{X}_i}=\left(\frac{\partial L}{\partial Y_i}\frac{\partial Y_i}{\partial \hat{X}_i}\right)^T=\left(\frac{\partial L}{\partial Y_i}\text{diag}(k)\right)^T=\frac{dL}{dY_i}\cdot k$

($\cdot$ is ė, elementwise multiplication)

② $\left(\frac{\partial \hat{X}_i}{\partial \sigma^2}\right)=-\frac{1}{2}(X_i-\mu)(\sigma^2+\varepsilon)^{-\frac{3}{2}}$

($\frac{\partial}{\partial}$ is elementwise)

$\frac{\partial \hat{X}_i}{\partial \sigma^2}=\text{diag}\left(\frac{\partial \hat{X}_i}{\partial \sigma^2}\right)$

$\therefore \frac{dL}{d\sigma^2}=\left(\frac{\partial L}{\partial \sigma^2}\right)^T=\left(\sum_i\frac{\partial L}{\partial \hat{X}_i}\frac{\partial \hat{X}_i}{\partial \sigma^2}\right)^T$

$=\sum_i\frac{dL}{d\hat{X}_i}\cdot\frac{\partial \hat{X}_i}{\partial \sigma^2}$

③ $\frac{\partial \hat{X}_i}{\partial \mu}=-\frac{1}{\sqrt{\sigma^2+\varepsilon}}$

$\frac{\partial \sigma^2}{\partial \mu}=-\frac{2}{m}\sum_i(X_i-\mu)=0$

$\therefore \frac{\partial L}{\partial \mu}=\sum_i\frac{\partial L}{\partial \hat{X}_i}\frac{\partial \hat{X}_i}{\partial \mu}+\frac{\partial L}{\partial \sigma^2}\frac{\partial \sigma^2}{\partial \mu}$

$\therefore \frac{dL}{d\mu}=\sum_i\frac{dL}{d\hat{X}_i}\cdot\frac{\partial \hat{X}_i}{\partial \mu}+\frac{dL}{d\sigma^2}\cdot\frac{\partial \sigma^2}{\partial \mu}$

$=\sum_i\frac{dL}{d\hat{X}_i}\cdot\frac{\partial \hat{X}_i}{\partial \mu}$

④ $\frac{\partial \mu}{\partial X_i}=\frac{1}{m}\begin{pmatrix}1\\ \vdots\\ 1\end{pmatrix}$ (i.e., $\begin{pmatrix}\frac{1}{m}\\ \frac{1}{m}\\ \vdots\\ \frac{1}{m}\end{pmatrix}$)

$\frac{\partial \sigma^2}{\partial X_i}=\frac{2}{m}(X_i-\mu)$

$\frac{\partial \hat{X}_i}{\partial X_i}=\frac{1}{\sqrt{\sigma^2+\varepsilon}}$

$\therefore \frac{\partial L}{\partial X_i}=\frac{\partial L}{\partial \hat{X}_i}\frac{\partial \hat{X}_i}{\partial X_i}+\frac{\partial L}{\partial \sigma^2}\frac{\partial \sigma^2}{\partial X_i}+\frac{\partial L}{\partial \mu}\frac{\partial \mu}{\partial X_i}$

$\therefore \frac{dL}{dX_i}=\frac{dL}{d\hat{X}_i}\frac{\partial \hat{X}_i}{\partial X_i}+\frac{dL}{d\sigma^2}\frac{\partial \sigma^2}{\partial X_i}+\frac{dL}{d\mu}\frac{\partial \mu}{\partial X_i}$

⑤ $\frac{dL}{dk}=\sum\frac{dL}{dY_i}\cdot\hat{X}_i$

$\frac{dL}{db}=\sum\frac{dL}{dY_i}$

⑥ $\frac{dL}{dX_i}=\frac{1}{m}\frac{k}{\sqrt{\sigma^2+\varepsilon}}\left(m\frac{dL}{dY_i}-\hat{X}_i\frac{dL}{dk}-\frac{dL}{db}\right)$

$\frac{dL}{dX}=\frac{1}{m}\frac{k}{\sqrt{\sigma^2+\varepsilon}}\cdot\left(m\frac{dL}{dY}-\hat{X}\cdot\frac{dL}{dk}-\frac{dL}{db}\right)$ ($\cdot$ is ė)

(here $\hat{X}=(\hat{X}_1,\cdots,\hat{X}_m)$)